Pandita's Aether-DCTWIN

Technology Stack & System Architecture Reference

1. End-to-End Architecture

LAYER 1 • FIELD
Facility Meters
BACnet/IP, Modbus TCP, SNMPv3, Sparkplug B & CSV/JSON imports.

→

LAYER 2 • API
FastAPI & WS
Async REST, 1.5s WebSocket push & YAML site blueprints.

→

LAYER 3 • AI & CORE
5-Layer AI & Physics
RLS calibration, IsolationForest, SciPy MPC, GraphRAG & LLM tools.

→

LAYER 4 • WEB & BMS
React + Three.js
3D WebGL floor, 10k-GPU planner, Copilot & BACnet fallback.

> **Facility-Only Boundary & AI Safety Rule:** Zero agents on tenant servers. LLMs orchestrate read-only tools and explain data—they never write setpoints directly; deterministic physics & constrained MPC enforce ASHRAE safety limits.

2. Frontend & 3D Visualization Stack

TECHNOLOGY	HOW IT IS USED IN AETHER-DCTWIN
React 18 & TypeScript	Strongly typed single-page workspace (`Overview` , `Inspect` , `Plan` , `Health` , `ROI` , `Copilot`) with live WebSocket state sync.
Three.js & React Three Fiber	Hardware-accelerated 3D WebGL data hall (`@react-three/fiber` , `@react-three/drei`) rendering rack heatmaps, overhead liquid coolant loops, airflow plumes, and click-to-inspect floor equipment.
Vite 5 & Dynamic Plugins	Fast production bundling plus conflict-free UI plugin discovery (`import.meta.glob`) so engineers add feature tabs independently.
Tailwind CSS & Recharts	Clean Dark/Light operational design system with tabular numeric alignment (`tabular-nums`) and interactive PUE / 30-minute forecast charts.

3. Backend, Streaming & Data Historian Stack

TECHNOLOGY	HOW IT IS USED IN AETHER-DCTWIN
Python 3.11+, FastAPI, Uvicorn	High-throughput async REST API, automatic OpenAPI/Swagger documentation (`/docs`), and real-time `/ws/telemetry` push server.
Pydantic v2 & PyYAML	Validates all API contracts and loads declarative facility blueprints (`configs/sites/*.yaml`) so any row/rack/CDU/tariff layout runs without code edits.
DuckDB & Apache Parquet	Embedded columnar historian (`duckdb` + `pyarrow`) storing 24-hour facility snapshots for sub-10ms DVR time-travel scrubbing and SQL analytics—zero external DB setup.
NetworkX	Models the directed electrical & cooling topology graph (`UPS → PDU → Rack ← CDU / CRAC`) for blast-radius checks and maintenance checklists.

4. What AI We Use — 5-Layer Hybrid AI & Physics Stack

1. Multi-Model Agentic LLMs

OpenRouter • Ollama

Supports `Claude Sonnet`, `Gemini 2.5 Pro`, `GPT-4o`, or air-gapped on-prem `Llama 3.3 70B` / `DeepSeek-R1` via structured JSON tool-calling across 7 live telemetry, SQL & simulation tools (zero hallucinated numbers).

2. Topology-Aware GraphRAG

NetworkX + SOP Engine

Traverses the live `UPS → PDU → Rack ← CDU/CRAC` dependency graph to collapse cascading rack alerts into 1 root cause, compute N+1 blast radius, and synthesize ASHRAE/LOTO `MOP` checklists.

3. Physics-Guided Anomaly ML

scikit-learn + State-Space

Fuses multivariate `IsolationForest` outlier scoring over electro-thermal ratios with energy-balance residuals and `5/15/30m` trajectory forecasting to distinguish fan trips or bypass leaks from normal AI bursts.

4. Online Self-Calibration AI

NumPy (RLS / Kalman)

Recursive Least Squares covariance filter continuously learns zone thermal resistance (`R_th`) and mass (`C_th`) from standard PDU/CDU meters—catching strainer and heat-exchanger fouling days early.

5. Safe Constrained Control (MPC)

SciPy SLSQP + CBF

Solves 30-minute cooling & UPS battery peak-shaving trajectories via sequential quadratic programming inside hard Control Barrier Functions (`ASHRAE ≤27°C`, dew-point guard, `≤4%/min` VFD slew).

6. First-Principles & 10k-GPU Engine

backend/domain/

Deterministic psychrometrics, cubic affinity laws, 3-phase AC phasors, and multi-MW `GB200 NVL72` / `HGX H200` cluster sizing (87/13 liquid split, 415V busways, <100ms BESS step-load buffers).

5. Field Integration, Team Modularity & Deployment

Field Protocol Safety

- **Pluggable Adapter Layer:** Clean interface separating simulation from live BACnet/IP, Modbus TCP, SNMPv3 & Sparkplug B gateways.
- **Automatic BMS Fallback:** Supervisory writes use BACnet Slot 8 and clear cleanly back to local Slot 16 defaults whenever Auto-Pilot is turned off.
- **BYOD CSV/JSON Mapper:** Instant offline evaluation on customer historian exports.

Parallel Team Velocity

- **Zero-Conflict Plugins:** Auto-mounted FastAPI routers (`api/features/*.py`) and React tabs let engineers build features without touching shared files.
- **Config-Driven Sites:** Customer topologies, equipment ratings & utility tariffs live in declarative YAML.
- **SWE Domain Onboarding:** Built-in Critical Facilities Rosetta Stone for non-MEP software engineers.

CI/CD & Deployment

- **Automated Guardrails:** `Pytest + GitHub Actions` CI verify physics laws, multi-site configs, and strict zero-server-access rules on every PR.
- **Single-Artifact Runtime:** FastAPI serves both the REST/WS backend and compiled React/WebGL bundle.
- **Container Ready:** Ships with `Dockerfile`, `docker-compose.yml`, and 1-command local launcher.